

Qualification Pack



Seasoning and Chemical Treatment Assistant (Woodware)

QP Code: HCS/Q6801

Version: 2.0

NSQF Level: 2.5

Handicrafts and Carpet Sector Skill Council || OCF, Plot No. 2, Pocket 9, Sector B, Vasant Kunj
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HCS/Q6801: Seasoning and Chemical Treatment Assistant (Woodware)

Brief Job Description

The individual is responsible for chemical treatment and seasoning of wood preparing it for further processing and works under supervision. The individual loads and unloads the chemical and heat treatment chambers, ensures correct maintenance of pressure and temperature, operates these chambers and is responsible for ensuring safe working. The person possesses knowledge of various wood species and relevant wood treatment.

Personal Attributes

The job requires the operator to be fit to lift logs and move or transport them. The person needs to be prepared for manual work and have an attentive and alert mind.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [HCS/N6608: Use basic health and safety practices in a wood working environment](#)
2. [HCS/N6801: Carry out chemical treatment of wood as per requirement](#)
3. [HCS/N6802: Carry out seasoning processes on wood](#)
4. [DGT/VSQ/N0101: Employability Skills \(30 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Woodware)
Occupation	Seasoning and Chemical Treatment
Country	India
NSQF Level	2.5
Credits	11
Aligned to NCO/ISCO/ISIC Code	NCO-2015/7521.9900

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Minimum Educational Qualification & Experience	9th grade pass OR 8th grade pass and pursuing continuous schooling OR 8th grade pass with 1 Year of experience OR 5th grade pass with 4 Years of experience OR Ability to read and write with 5 Years of experience OR Previous relevant Qualification of NSQF Level (2) with 6 Months of experience OR Previous relevant Qualification of NSQF Level (1) with 1-2 Years of experience
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	31/08/2026
NSQC Approval Date	31/08/2023
Version	2.0
Reference code on NQR	QG-2.5-HC-00858-2023-V2-HCSSC
NQR Version	2.0

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HCS/N6608: Use basic health and safety practices in a wood working environment

Description

This unit covers health and safety procedures to be followed as required in a woodworking environment in the handicrafts and carpet sector. This include following health and safety measures as may be suggested by the organizations policies and documents.

Scope

The scope covers the following :

- general health and safety
- fire safety
- physical safety
- chemical safety
- electrical safety
- incident response

Elements and Performance Criteria

General health and safety

To be competent, the user/individual on the job must be able to:

- PC1.** use protective clothing/equipment for specific tasks and work conditions Protective clothing: Leather or asbestos gloves, aprons, trousers, footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, particle masks, glasses/goggles/visors Equipment: Hand shields, machine guards, residual current devices, shields, dust sheets, respirator
- PC2.** state the name and location of people responsible for health and safety in the workplace
- PC3.** state the names and location of documents that refer to health and safety in the workplace
- PC4.** Identify job site of hazardous work and state possible causes of risk or accident in the workplace Hazards: Sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.);hazardous substances(chemicals, gas, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tools and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages,etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness) understand various health and safety measures that are important in the organizations functioning

Fire safety

To be competent, the user/individual on the job must be able to:

- PC5.** Carry out jobs which cause sparking in a safe manner, creating sufficient distance between flammable items and the sparking

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- PC6.** Use fire extinguishers appropriately in accordance to the type of fire correctlyTypes of fires:
Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.;
Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents)
- PC7.** Demonstrate rescue techniques applied during fire hazard
- PC8.** Demonstrate good housekeeping in order to prevent fire hazards
- PC9.** Demonstrate the correct use of a fire extinguisher

Physical Safety

To be competent, the user/individual on the job must be able to:

- PC10.** Inspect for faults; set up and safely use steps and ladders in general useLadder faults:
Corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/ unfixed nuts or bolts, etc. Ladders set up: Firm/level base, clip/lash down, leaning at the correct angle, etc
- PC11.** Work safely in and around trenches, elevated places and confined areas
- PC12.** Lift heavy objects safely using correct procedures
- PC13.** Ensure that the machine guards are in place in order to ascertain safety when working with machines with sharp edges, blades, abrasive materials, etc.

Chemical Safety

To be competent, the user/individual on the job must be able to:

- PC14.** Identify different types of chemicals that are used in the wood finishing and treatment
- PC15.** Use chemicals as per prescribed levels for any type of chemical treatment and finishing of the wood
- PC16.** explain to others the harmful effects of any chemical in use in excess or in appropriate combination
- PC17.** Use safety equipments while using sprays and paints
- PC18.** Ensure adherence of proper limits while maintaining the edible level of colors and paints when used on toys and dolls

Electrical safety

To be competent, the user/individual on the job must be able to:

- PC19.** Ensure all electrical appliances are used in line with instructions and with the appropriate attachments in line with manufacturers guidelines
- PC20.** Spot and repair any frayed wires and insulation using appropriate and approved measures like insulation taping, replacing wires, etc.
- PC21.** Ensure all naked wires are covered with insulation
- PC22.** Wear insulated footwear while working with electrical appliances or equipment

Incident response

To be competent, the user/individual on the job must be able to:

- PC23.** Demonstrate how to free a person from electrocution
- PC24.** Administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.

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- PC25.** Demonstrate basic techniques of bandaging
- PC26.** respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments
- PC27.** Perform and organize required responses for loss minimization or rescue activity during an accident in real or simulated environments
- PC28.** Administer first aid responses to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases
- PC29.** Demonstrate the artificial respiration and the CPR Process
- PC30.** Participate in emergency procedures
Emergency procedures: Raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work
- PC31.** Complete a written accident/incident report or dictate a report to another person, and send report to person responsible
Incident Report includes details of: Name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified
- PC32.** Demonstrate correct methods to move injured people and others during an emergency

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Organization's policies on safety measures
- KU2.** importance of the individual's role and its importance with respect to daily operations of the organization
- KU3.** Formal reporting and operating structure of the organization
- KU4.** Documentation and recording policies of the organization
- KU5.** The extent and categories of the different market areas targeted by the organization
- KU6.** Type and properties of different wood quality
- KU7.** Quality of wood and other raw material used to achieve the right finish
- KU8.** Safety equipment and precautions to be taken
- KU9.** Meaning of hazards and risks
- KU10.** health and safety hazards commonly present in the work environment and related precautions
- KU11.** possible causes of risk and accident
Possible causes of risk and accident: Physical actions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness)
- KU12.** Methods of accident prevention
Methods of accident prevention: Training in health and safety procedures; using health and safety procedures; use of equipment and working practices (such as safe carrying procedures); safety notices, advice; instruction from colleagues and supervisors
- KU13.** Safe working practices when working with tools and machines
- KU14.** Safe working practices while working at various hazardous sites
- KU15.** safe working practices while working at various hazardous sites
- KU16.** where to find all the general health and safety equipment in the workplace
- KU17.** various dangers associated with the use of electrical equipment

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- KU18.** hazards and risks associated with working with wood Hazards and risks: Flammable nature of wood and wood shavings, weight of wood, sharp edges and wooden splinters, breakages, etc.
- KU19.** preventative and remedial actions to be taken in the case of exposure to toxic materials Exposure: Ingested, contact with skin, inhaled Preventative action: Ventilation, masks, protective clothing/ equipment; Remedial action: immediate first aid, report to supervisor Toxic materials: Solvents, lead, acid
- KU20.** importance of using protective clothing/equipment while working
- KU21.** precautionary activities to prevent the fire accident
- KU22.** various causes of fire Causes of fires: Heating of metal; spontaneous ignition; sparking; electrical heating; loose fires (smoking, welding, etc.); chemical fires; etc.
- KU23.** techniques of using the different fire extinguishers
- KU24.** different methods of extinguishing fire
- KU25.** different materials used for extinguishing fire Materials: sand, water, foam, CO₂, dry powder

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read the organization's health and safety policy documents or notifications in the local language using assistance with difficult words, sentences and passages
- GS2.** read the safety manuals and/or instructions of various machines and tools being used in the organization using assistance with difficult words and passages
- GS3.** read and comprehend basic content to read labels, charts, permissible levels of use, signages
- GS4.** Write simple messages for communicating health and safety warnings or instructions in the local language clearly and coherently
- GS5.** Write quantities and names for materials and equipments for requisitioning, issuing and receiving purposes
- GS6.** Speak clearly in the local language to communicate basic health and safety instructions effectively
- GS7.** Use appropriate protocol (titles, courtesies, etc.) to communicate with different levels of co-workers
- GS8.** Display active listening skills while interacting with co-workers and other in the workplace
- GS9.** Give instructions to the workforce regarding the safe storage and handling of the chemicals and other related material in the organization
- GS10.** Consider time, safety and quality as parameters to escalate issues to supervisor or modify work schedules
- GS11.** Use correct sequencing and time planning techniques to carry out health and safety tasks in the time allocated correctly and effectively
- GS12.** Use checklists to ensure all health and safety actions are taken before, during and after work operations
- GS13.** Work with co-workers and supervisor to resolve any issues that threaten disruption or increase risk of health and safety

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- GS14.** Anticipate and work towards reducing the possible risks that can emerge in a woodworking environment
- GS15.** Improvise practices and processes using safe and acceptable methods, in discussions with the supervisor in order to ensure improved safety and health practices
- GS16.** Diagnose any problem leading to production delay or disruption
- GS17.** Diagnose any problem in the machine handling or chemical handling while working and storage
- GS18.** Report any concern to the relevant authorities

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>General health and safety</i>	3	8	-	-
PC1. use protective clothing/equipment for specific tasks and work conditions Protective clothing: Leather or asbestos gloves, aprons, trousers, footwear, helmets/hard hats, cap and shoulder covers, ear defenders/plugs, safety boots, particle masks, glasses/goggles/visors Equipment: Hand shields, machine guards, residual current devices, shields, dust sheets, respirator	-	2	-	-
PC2. state the name and location of people responsible for health and safety in the workplace	1	2	-	-
PC3. state the names and location of documents that refer to health and safety in the workplace	1	2	-	-
PC4. Identify job site of hazardous work and state possible causes of risk or accident in the workplace Hazards: Sharp edged and heavy tools; heated metals; oxyfuel and gas cylinders; hazardous surfaces(sharp, slippery, uneven, chipped, broken, etc.);hazardous substances(chemicals, gas, fumes, dust, etc.); physical hazards(working at heights, large and heavy objects and machines, sharp and piercing objects, tolls and machines, intense light, load noise, obstructions in corridors, by doors, blind turns, noise, over stacked shelves and packages,etc.) electrical hazards (power supply and points, loose and naked cables and wires, electrical machines and appliances, etc.) Possible causes of risk and accident: physical actions; reading; listening to and giving instructions; inattention; sickness and incapacity (such as drunkenness); health hazards (such as untreated injuries and contagious illness)understand various health and safety measures that are important in the organizations functioning	1	2	-	-
<i>Fire safety</i>	1	13	-	-
PC5. Carry out jobs which cause sparking in a safe manner, creating sufficient distance between flammable items and the sparking	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC6. Use fire extinguishers appropriately in accordance to the type of fire correctly Types of fires: Class A: eg. ordinary solid combustibles, such as wood, paper, cloth, plastic, charcoal, etc.; Class B: flammable liquids and gases, such as gasoline, propane, diesel fuel, tar, cooking oil, and similar substances; Class C: eg. electrical equipment such as appliances, wiring, breaker panels, etc. (These categories of fires become Class A, B, and D fires when the electrical equipment that initiated the fire is no longer receiving electricity); Class D: combustible metals such as magnesium, titanium, and sodium (These fires burn at extremely high temperatures and require special suppression agents)	-	2	-	-
PC7. Demonstrate rescue techniques applied during fire hazard	-	3	-	-
PC8. Demonstrate good housekeeping in order to prevent fire hazards	-	3	-	-
PC9. Demonstrate the correct use of a fire extinguisher	-	3	-	-
<i>Physical Safety</i>	3	8	-	-
PC10. Inspect for faults; set up and safely use steps and ladders in general use Ladder faults: Corrosion of metal components, deterioration, splits and cracks timber components, imbalance, loose rungs, missing/unfixed nuts or bolts, etc. Ladders set up: Firm/level base, clip/lash down, leaning at the correct angle, etc	1	3	-	-
PC11. Work safely in and around trenches, elevated places and confined areas	-	2	-	-
PC12. Lift heavy objects safely using correct procedures	1	1	-	-
PC13. Ensure that the machine guards are in place in order to ascertain safety when working with machines with sharp edges, blades, abrasive materials, etc.	1	2	-	-
<i>Chemical Safety</i>	3	11	-	-
PC14. Identify different types of chemicals that are used in the wood finishing and treatment	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC15. Use chemicals as per prescribed levels for any type of chemical treatment and finishing of the wood	1	3	-	-
PC16. explain to others the harmful effects of any chemical in use in excess or in appropriate combination	-	2	-	-
PC17. Use safety equipments while using sprays and paints	-	2	-	-
PC18. Ensure adherence of proper limits while maintaining the edible level of colors and paints when used on toys and dolls	1	2	-	-
<i>Electrical safety</i>	2	12	-	-
PC19. Ensure all electrical appliances are used in line with instructions and with the appropriate attachments in line with manufacturers guidelines	1	3	-	-
PC20. Spot and repair any frayed wires and insulation using appropriate and approved measures like insulation taping, replacing wires, etc.	1	3	-	-
PC21. Ensure all naked wires are covered with insulation	-	3	-	-
PC22. Wear insulated footwear while working with electrical appliances or equipment	-	3	-	-
<i>Incident response</i>	8	28	-	-
PC23. Demonstrate how to free a person from electrocution	-	3	-	-
PC24. Administer appropriate first aid to victims where required eg. in case of bleeding, burns, choking, electric shock, poisoning etc.	1	2	-	-
PC25. Demonstrate basic techniques of bandaging	1	1	-	-
PC26. respond promptly and appropriately to an accident situation or medical emergency in real or simulated environments	-	3	-	-
PC27. Perform and organize required responses for loss minimization or rescue activity during an accident in real or simulated environments	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC28. Administer first aid responses to victims in case of a heart attack or cardiac arrest due to electric shock, before the arrival of emergency services in real or simulated cases	1	3	-	-
PC29. Demonstrate the artificial respiration and the CPR Process	1	3	-	-
PC30. Participate in emergency procedures Emergency procedures: Raising alarm, safe/efficient, evacuation, correct means of escape, correct assembly point, roll call, correct return to work	1	3	-	-
PC31. Complete a written accident/incident report or dictate a report to another person, and send report to person responsible Incident Report includes details of: Name, date/time of incident, date/time of report, location, environment conditions, persons involved, sequence of events, injuries sustained, damage sustained, actions taken, witnesses, supervisor/manager notified	1	4	-	-
PC32. Demonstrate correct methods to move injured people and others during an emergency	1	4	-	-
NOS Total	20	80	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N6608
NOS Name	Use basic health and safety practices in a wood working environment
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Woodware)
Occupation	Merchandising, Pre-Production, Assembling, Finishing
NSQF Level	2.5
Credits	2
Version	2.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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HCS/N6801: Carry out chemical treatment of wood as per requirement

Description

This unit involves carrying out the chemical treatment of wood. It also covers knowledge of wood species and wood treatment as per species and thickness of wood.

Scope

The scope covers the following :

- General health and safety
- fire safety
- physical safety
- chemical safety
- electrical safety
- incident response

Elements and Performance Criteria

Prepare for chemical treatment of wood

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of wood species
- PC2.** stack wooden pieces safely and as per organisational standards, in designate area
- PC3.** transport wooden planks and pieces safely from one place to another, as required, using manual lifting and trolleys
- PC4.** load chemicals in the chemical seasoning chamber by operating the valves correctly
- PC5.** load and unload wooden planks into the chamber, to ensure optimum utilisation and safety

Remove bugs through chemical treatment of wooden logs and planks

To be competent, the user/individual on the job must be able to:

- PC6.** diagnose the presence or absence of bugs, termites, fungus attack etc. in the wood, using visual inspection
- PC7.** identify and segregate the pieces that are highly damaged and are beyond the treatment point, and get duly verified by supervisor
- PC8.** diagnose the intensity of chemical treatment required by the individual pieces under guidance from supervisor
- PC9.** monitor the gauges on the chemical seasoning chamber to ensure correct pressure

Check pieces of wood post chemical treatment

To be competent, the user/individual on the job must be able to:

- PC10.** check every individual piece after the chemical treatment for correct exposure and any gaps in treatment
- PC11.** ensure that the chemical treatment has served the purpose and the wood log/plank is free of bugs, termites , fungus etc.
- PC12.** check and send the complete wood log batch for further processing

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Work safely

To be competent, the user/individual on the job must be able to:

- PC13.** use personal protective equipment (ppe) as required for the jobPPE: eg. gloves, protective footwear, aprons, goggles, breathing mask, etc.
- PC14.** handle the chemicals safely and efficiently, without spills and in appropriate containers
- PC15.** avoid any health and safety hazard due to mishandling of the chemicals
- PC16.** keep work area clean and orderly as per organizational standards
- PC17.** apply correct lifting techniques while handling heavy loads
- PC18.** identify and report any malfunction of the seasoning chamber to the supervisor after taking necessary safety precautions
- PC19.** communicate clearly in a timely manner to co-workers on matters related to health and safety

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organization's policies on safety measures, prescribed level of chemicals to be used and quality standards accepted
- KU2.** work flow involved in chemical treatment
- KU3.** importance of the individual's role in the production process of the organization
- KU4.** the formal reporting and operating structure of the organization
- KU5.** type and properties of different wood species and quality
- KU6.** the need and importance for wood treatment and the process
- KU7.** different chemicals used in wood treatment
- KU8.** quantities, duration and sequence of chemical application
- KU9.** process of chemical treatment of wood
- KU10.** various relevant treatment processes available in the industry
- KU11.** various technical developments in the field of wood treatment
- KU12.** defects and possible errors involved in the treatment process
- KU13.** quality of wood and other raw material used to achieve the right output
- KU14.** personal protective safety equipments, its need and correct use
- KU15.** precautions to be taken while working with chemicals
- KU16.** quality standards to be followed
- KU17.**
 - different levels of temperature and time required for different wood qualities and moisture
 - level
- KU18.** process to use a moisture meter
- KU19.** safe lifting, stacking and moving techniques for handling heavy loads
- KU20.** importance of temperature, pressure and time in the chemical treatment process
- KU21.** importance of correct stacking, loading and unloading, and its impact on quality and productivity
- KU22.** what is productivity and its importance for the company success

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Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read basic terms and terminologies to accurately interpret supervisor instructions in the local language
- GS2.** read and interpret accurate information from various relevant logs and records
- GS3.** read and interpret safety warnings, signage and labels
- GS4.** write the raw material requirement for operational requirements with complete detailing in the local language
- GS5.** write basic numbers, quantities and messages for operational requirements in the local language
- GS6.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the job, work plans and other associated requirements
- GS7.** give clear instructions to co-workers about the type of output required, health & safety matters, and answer queries
- GS8.** display active listening skills while interacting with co-workers and others in the workplace
- GS9.** identify and list variables in the operational process that impact achievement and quality
- GS10.** state rules in order to make adjustments to work plan based on variables
- GS11.** consider time, safety and quality as parameters to escalate issues to supervisor or modify work schedules
- GS12.** use correct sequencing and time planning techniques to carry out operational tasks in the time allocated correctly and effectively
- GS13.** use checklists to ensure all actions are taken before, during and after work operations as required according to company procedures and job requirements
- GS14.** link the importance of following processes and specifications and connecting it to quality and ultimately shaping customer experience
- GS15.** explain the importance of customer satisfaction for company and personal success
- GS16.** explain the concept of internal and external customers
- GS17.** work with co-workers and supervisor to resolve any issues that threaten disruption, increase risk, cause delays or under-achievement of quality and targets as per the planned schedule
- GS18.** anticipate and work towards reducing the possible errors that can emerge in the chemical treatment process
- GS19.** improvise on the work process using safe and acceptable methods in order to ensure achievement of targets or improvements in productivity and output
- GS20.** identify any problem leading to production output delay or disruption.
- GS21.** report any concern to the relevant authorities where there is risk to men, materials, tools and equipment

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Prepare for chemical treatment of wood</i>	5	22	-	-
PC1. identify different types of wood species	1	4	-	-
PC2. stack wooden pieces safely and as per organisational standards, in designate area	2	4	-	-
PC3. transport wooden planks and pieces safely from one place to another, as required, using manual lifting and trolleys	-	4	-	-
PC4. load chemicals in the chemical seasoning chamber by operating the valves correctly	2	6	-	-
PC5. load and unload wooden planks into the chamber, to ensure optimum utilisation and safety	-	4	-	-
<i>Remove bugs through chemical treatment of wooden logs and planks</i>	4	19	-	-
PC6. diagnose the presence or absence of bugs, termites, fungus attack etc. in the wood, using visual inspection	-	4	-	-
PC7. identify and segregate the pieces that are highly damaged and are beyond the treatment point, and get duly verified by supervisor	1	4	-	-
PC8. diagnose the intensity of chemical treatment required by the individual pieces under guidance from supervisor	2	6	-	-
PC9. monitor the gauges on the chemical seasoning chamber to ensure correct pressure	1	5	-	-
<i>Check pieces of wood post chemical treatment</i>	3	11	-	-
PC10. check every individual piece after the chemical treatment for correct exposure and any gaps in treatment	1	4	-	-
PC11. ensure that the chemical treatment has served the purpose and the wood log/plank is free of bugs, termites, fungus etc.	1	4	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check and send the complete wood log batch for further processing	1	3	-	-
<i>Work safely</i>	8	28	-	-
PC13. use personal protective equipment (ppe) as required for the jobPPE: eg. gloves, protective footwear, aprons, goggles, breathing mask, etc.	1	4	-	-
PC14. handle the chemicals safely and efficiently, without spills and in appropriate containers	1	5	-	-
PC15. avoid any health and safety hazard due to mishandling of the chemicals	1	4	-	-
PC16. keep work area clean and orderly as per organizational standards	2	3	-	-
PC17. apply correct lifting techniques while handling heavy loads	1	3	-	-
PC18. identify and report any malfunction of the seasoning chamber to the supervisor after taking necessary safety precautions	1	4	-	-
PC19. communicate clearly in a timely manner to co-workers on matters related to health and safety	1	5	-	-
NOS Total	20	80	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	HCS/N6801
NOS Name	Carry out chemical treatment of wood as per requirement
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Woodware)
Occupation	Pre-Production
NSQF Level	2.5
Credits	5
Version	2.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

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HCS/N6802: Carry out seasoning processes on wood

Description

This unit performs the seasoning of wood for individual pieces of wooden planks.

Scope

The scope covers the following :

- handle seasoning machinery and other equipments
- check wooden pieces after seasoning treatment for further processing
- work safely

Elements and Performance Criteria

Handle seasoning machinery and other equipments

To be competent, the user/individual on the job must be able to:

- PC1.** identify different types of wood qualities and species
- PC2.** stack wooden pieces safely and as per organisational standards, in designate area
- PC3.** transport wooden planks and pieces safely from one place to another, as required, using manual lifting and trolleys
- PC4.** load and unload wooden planks into the chamber or kiln, to ensure optimum utilisation and safety; verify it with the supervisor
- PC5.** load fuel or wood chippings to fire the chamber in a safe manner in required quantities, ensuring sufficient stock for replenishing
- PC6.** take all the necessary safety and precautionary measures as prescribed by the organization

Perform seasoning treatment of wooden planks

To be competent, the user/individual on the job must be able to:

- PC7.** diagnose the level of moisture present in the wood using a moisture meter
- PC8.** reject the pieces that are highly damp and are beyond the seasoning point, verify with supervisor
- PC9.** fire up the chamber after ensuring it is properly locked
- PC10.** maintain desired temperature and pressure by replenishing fuel or wood chippings
- PC11.** perform the seasoning treatment of the individual wooden pieces keeping in mind the required time and temperature for every individual piece

Check wooden pieces after seasoning treatment for further processing

To be competent, the user/individual on the job must be able to:

- PC12.** check every individual piece after the seasoning treatment where the quantity is low and use appropriate random sampling techniques to check when quantities are large
- PC13.** use a moisture meter accurately for checking moisture in wooden pieces
- PC14.** ensure that the seasoning treatment has served the purpose and the wood log is free of moisture and other defects
- PC15.** diagnose and re-treat the wooden logs that need further seasoning treatment

Qualification Pack

PC16. check and send the complete wood log batch for further processing after supervisor approval

Work safely

To be competent, the user/individual on the job must be able to:

PC17. handle the tools and equipment safely and efficiently

PC18. use personal protective equipment (ppe) as required for the job PPE: eg. gloves, protective footwear, aprons, goggles, breathing mask, etc

PC19. keep work area clean and orderly as per organizational standards

PC20. apply correct lifting techniques while handling heavy loads

PC21. take precautions to ensure flammable items are not placed near the heat sources

PC22. identify and report any malfunction of the seasoning chamber to the supervisor after taking necessary safety precautions

PC23. communicate clearly in a timely manner to co-workers on matters related to health and safe

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. organization's policies on safety measures, prescribed level of chemicals to be used and quality standards accepted

KU2. work flow involved in seasoning treatment

KU3. importance of the individual's role in the production process of the organization

KU4. the formal reporting and operating structure of the organization

KU5. documentation and recording policies of the organization

KU6. type and properties of different wood quality

KU7. importance of wood seasoning

KU8. wood seasoning process

KU9. different functions and operations of the seasoning equipment

KU10. various relevant treatment processes available in the industry

KU11. various technical developments in the field of wood treatment

KU12. various kinds of raw materials used in the treatment process

KU13. defects and possible errors involved in the treatment process

KU14. quality of wood and other raw material used to achieve the right output

KU15. personal protective safety equipments, its need and correct use

KU16. quality standards to be followed

KU17. different levels of temperature and time required for different wood qualities and moisture level

KU18. process to use a moisture meter

KU19. safe lifting, stacking and moving techniques for handling heavy loads

KU20. importance of temperature, pressure and time in the chemical treatment process

KU21. importance of correct stacking, loading and unloading, and its impact on quality and productivity

KU22. what is productivity and its importance for the company's success

Qualification Pack

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read basic terms and terminologies to accurately interpret supervisor instructions in the local language
- GS2.** read and interpret accurate information from various relevant logs and records
- GS3.** read and interpret safety warnings, signage and labels
- GS4.** write the raw material requirement for operational requirements with complete detailing in the local language
- GS5.** write basic numbers, quantities and messages for operational requirements in the local language
- GS6.** interact with the supervisor appropriately (correct protocol and manner of speaking) in order to understand the basic requirements of the job, work plans and other associated requirements
- GS7.** give clear instructions to co-workers about the type of output required, health & safety matters, and answer queries
- GS8.** display active listening skills while interacting with co-workers and other in the workplace
- GS9.** identify and list variables in the operational process that impact achievement and quality
- GS10.** state rules in order to make adjustments to work plan based on variables
- GS11.** consider time, safety and quality as parameters to escalate issues to supervisor or modify work schedules
- GS12.** use correct sequencing and time planning techniques to carry out operational tasks in the time allocated correctly and effectively
- GS13.** use checklists to ensure all actions are taken before, during and after work operations as required according to company procedures and job requirements
- GS14.** link the importance of following processes and specifications and connecting it to quality and ultimately shaping customer experience
- GS15.** explain the importance of customer satisfaction for company and personal success
- GS16.** explain the concept of internal and external customers
- GS17.** work with co-workers and supervisor to resolve any issues that threaten disruption, increase risk, cause delays or under-achievement of quality and targets as per the planned schedule
- GS18.** anticipate and work towards reducing the possible errors that can emerge in the chemical treatment process
- GS19.** improvise on the work process using safe and acceptable methods in order to ensure achievement of targets or improvements in productivity and output
- GS20.** identify any problem leading to production output delay or disruption.
- GS21.** report any concern to the relevant authorities where there is risk to men, materials, tools and equipment or achievement

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Handle seasoning machinery and other equipments</i>	6	24	-	-
PC1. identify different types of wood qualities and species	1	4	-	-
PC2. stack wooden pieces safely and as per organisational standards, in designate area	2	3	-	-
PC3. transport wooden planks and pieces safely from one place to another, as required, using manual lifting and trolleys	-	4	-	-
PC4. load and unload wooden planks into the chamber or kiln, to ensure optimum utilisation and safety; verify it with the supervisor	1	4	-	-
PC5. load fuel or wood chippings to fire the chamber in a safe manner in required quantities, ensuring sufficient stock for replenishing	-	5	-	-
PC6. take all the necessary safety and precautionary measures as prescribed by the organization	2	4	-	-
<i>Perform seasoning treatment of wooden planks</i>	4	18	-	-
PC7. diagnose the level of moisture present in the wood using a moisture meter	2	3	-	-
PC8. reject the pieces that are highly damp and are beyond the seasoning point, verify with supervisor	-	4	-	-
PC9. fire up the chamber after ensuring it is properly locked	-	3	-	-
PC10. maintain desired temperature and pressure by replenishing fuel or wood chippings	1	3	-	-
PC11. perform the seasoning treatment of the individual wooden pieces keeping in mind the required time and temperature for every individual piece	1	5	-	-
<i>Check wooden pieces after seasoning treatment for further processing</i>	6	16	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC12. check every individual piece after the seasoning treatment where the quantity is low and use appropriate random sampling techniques to check when quantities are large	1	4	-	-
PC13. use a moisture meter accurately for checking moisture in wooden pieces	2	3	-	-
PC14. ensure that the seasoning treatment has served the purpose and the wood log is free of moisture and other defects	1	3	-	-
PC15. diagnose and re-treat the wooden logs that need further seasoning treatment	1	3	-	-
PC16. check and send the complete wood log batch for further processing after supervisor approval	1	3	-	-
<i>Work safely</i>	4	22	-	-
PC17. handle the tools and equipment safely and efficiently	-	3	-	-
PC18. use personal protective equipment (ppe) as required for the jobPPE: eg. gloves, protective footwear, aprons, goggles, breathing mask, etc	-	3	-	-
PC19. keep work area clean and orderly as per organizational standards	1	4	-	-
PC20. apply correct lifting techniques while handling heavy loads	1	3	-	-
PC21. take precautions to ensure flammable items are not placed near the heat sources	-	3	-	-
PC22. identify and report any malfunction of the seasoning chamber to the supervisor after taking necessary safety precautions	1	3	-	-
PC23. communicate clearly in a timely manner to co-workers on matters related to health and safe	1	3	-	-
NOS Total	20	80	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	HCS/N6802
NOS Name	Carry out seasoning processes on wood
Sector	Handicrafts and Carpet
Sub-Sector	Handicrafts (Woodware)
Occupation	Pre-Production
NSQF Level	2.5
Credits	3
Version	2.0
Last Reviewed Date	31/08/2023
Next Review Date	31/08/2026
NSQC Clearance Date	31/08/2023

Qualification Pack

DGT/VSQ/N0101: Employability Skills (30 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

PC1. understand the significance of employability skills in meeting the job requirements

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.

Basic English Skills

To be competent, the user/individual on the job must be able to:

PC4. speak with others using some basic English phrases or sentences

Communication Skills

To be competent, the user/individual on the job must be able to:

PC5. follow good manners while communicating with others

PC6. work with others in a team

Qualification Pack

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

PC7. communicate and behave appropriately with all genders and PwD

PC8. report any issues related to sexual harassment

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

PC9. use various financial products and services safely and securely

PC10. calculate income, expenses, savings etc.

PC11. approach the concerned authorities for any exploitation as per legal rights and laws

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

PC12. operate digital devices and use its features and applications securely and safely

PC13. use internet and social media platforms securely and safely

Entrepreneurship

To be competent, the user/individual on the job must be able to:

PC14. identify and assess opportunities for potential business

PC15. identify sources for arranging money and associated financial and legal challenges

Customer Service

To be competent, the user/individual on the job must be able to:

PC16. identify different types of customers

PC17. identify customer needs and address them appropriately

PC18. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC19. create a basic biodata

PC20. search for suitable jobs and apply

PC21. identify and register apprenticeship opportunities as per requirement

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use basic spoken English language

KU6. Do and dont of effective communication

KU7. inclusivity and its importance

KU8. different types of disabilities and appropriate communication and behaviour towards PwD

KU9. different types of financial products and services

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- KU10.** how to compute income and expenses
- KU11.** importance of maintaining safety and security in financial transactions
- KU12.** different legal rights and laws
- KU13.** how to operate digital devices and applications safely and securely
- KU14.** ways to identify business opportunities
- KU15.** types of customers and their needs
- KU16.** how to apply for a job and prepare for an interview
- KU17.** apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** communicate effectively using appropriate language
- GS2.** behave politely and appropriately with all
- GS3.** perform basic calculations
- GS4.** solve problems effectively
- GS5.** be careful and attentive at work
- GS6.** use time effectively
- GS7.** maintain hygiene and sanitisation to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. understand the significance of employability skills in meeting the job requirements	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC2. identify constitutional values, civic rights, duties, personal values and ethics and environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	1	3	-	-
PC3. explain 21st Century Skills such as Self-Awareness, Behavior Skills, Positive attitude, self-motivation, problem-solving, creative thinking, time management, social and cultural awareness, emotional awareness, continuous learning mindset etc.	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC4. speak with others using some basic English phrases or sentences	-	-	-	-
<i>Communication Skills</i>	1	1	-	-
PC5. follow good manners while communicating with others	-	-	-	-
PC6. work with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	1	-	-
PC7. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC8. report any issues related to sexual harassment	-	-	-	-
<i>Financial and Legal Literacy</i>	3	4	-	-
PC9. use various financial products and services safely and securely	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. calculate income, expenses, savings etc.	-	-	-	-
PC11. approach the concerned authorities for any exploitation as per legal rights and laws	-	-	-	-
<i>Essential Digital Skills</i>	4	6	-	-
PC12. operate digital devices and use its features and applications securely and safely	-	-	-	-
PC13. use internet and social media platforms securely and safely	-	-	-	-
<i>Entrepreneurship</i>	3	5	-	-
PC14. identify and assess opportunities for potential business	-	-	-	-
PC15. identify sources for arranging money and associated financial and legal challenges	-	-	-	-
<i>Customer Service</i>	2	2	-	-
PC16. identify different types of customers	-	-	-	-
PC17. identify customer needs and address them appropriately	-	-	-	-
PC18. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	1	3	-	-
PC19. create a basic biodata	-	-	-	-
PC20. search for suitable jobs and apply	-	-	-	-
PC21. identify and register apprenticeship opportunities as per requirement	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0101
NOS Name	Employability Skills (30 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	2
Credits	1
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Element/ Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each Element/ PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/ training center based on these criteria.
6. To pass the Qualification Pack assessment, every trainee should score the Recommended Pass % aggregate for the QP.
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

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Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
HCS/N6608.Use basic health and safety practices in a wood working environment	20	80	0	0	100	20
HCS/N6801.Carry out chemical treatment of wood as per requirement	20	80	0	0	100	20
HCS/N6802.Carry out seasoning processes on wood	20	80	0	0	100	20
DGT/VSQ/N0101.Employability Skills (30 Hours)	20	30	-	-	50	40
Total	80	270	-	-	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

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Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.